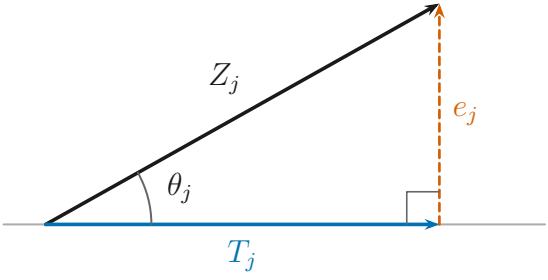
(a) Site level

Reliability as projection



$$T_j = \tau_j - \tau$$

$$Z_j = \hat{\tau}_j - \tau = T_j + e_j$$

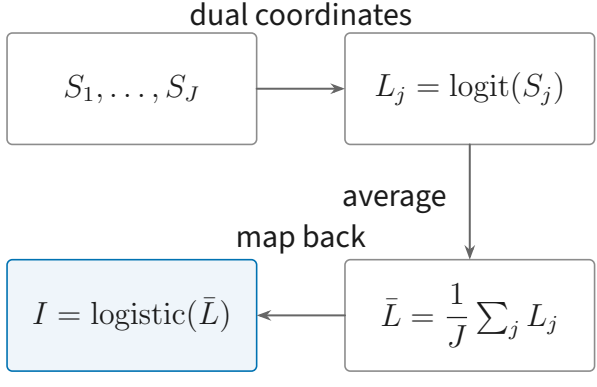
$$\mathbb{E}(e_j \mid T_j) = 0$$

$$S_j = \frac{\sigma^2}{\sigma^2 + v_j} = \cos^2 \theta_j$$

$$\omega_j = 1 - S_j = \sin^2 \theta_j$$

(b) Trial level

Informativeness as centroid



$$\text{logit}(S_j) = \ln \sigma^2 - \ln v_j$$

$$I = \frac{\sigma^2}{\sigma^2 + \text{GM}(v)}$$

Chosen first-argument KL centroid

$$H = \frac{1}{J} \sum_j \text{KL}\{\text{Bern}(I) \parallel \text{Bern}(S_j)\}$$